

HACKATHON PROBLEM STATEMENTS

Build. Break. Defend.

Two tracks: Game Development & Behavioral Security

Terminal Mafia (36 hrs) | **BioPrint: Behavioral Auth (36 hrs)**

Event 1 | Terminal Mafia

A local-hosted, multiplayer, terminal-based social deduction game

Duration: 36 hours

Hosting: Local device / LAN / hotspot

Theme: Terminal Mafia

Problem Statement

Build a multiplayer, terminal-based (CLI) social deduction game in the spirit of Mafia / Among Us. Players are assigned hidden roles (e.g., Mafia vs. Villagers, or a Betrayer among Cooperators), and the game must support a discussion/action phase, a voting or elimination mechanic, and distinct information visible only to each player based on their role.

The game must be playable entirely within the terminal (no GUI) and hosted locally: either on a single machine using multiple terminal windows, or across devices connected over the same local network or hotspot. No cloud deployment is required.

Core Requirements

- Supports 4+ players with hidden, role-based information (each player only sees what their role permits)
- A structured game loop: role assignment → phases (e.g., night/day, discuss/vote) → elimination → win condition
- Fully playable via terminal input/output: text-based only, no GUI frameworks
- Reasonably graceful handling of a player disconnecting or submitting invalid input mid-game
- A clear, unambiguous win condition for each side (e.g., Mafia eliminated / Mafia outnumbers Villagers)

Stretch Goals (optional, bonus credit)

- AI-controlled bot players to fill out a lobby when fewer humans are available
- Spectator mode for eliminated players
- Additional special roles (e.g., Detective, Doctor, Double Agent) beyond the basic two sides
- Match history / replay of role reveals and voting patterns at game end

Deliverables

Item	Description	Non-Submission Penalty
Git Repository	The team must share the Git Repository containing all the source code for the game along with a valid commit history and a ReadMe file	Disqualification
Executable File	Each Team must submit an executable file that can be directly loaded	-20 points
Demonstration Video	Each Team must submit a demonstration 3–5-minute demonstration video of their game played by multiple individuals	-10 points

Judging Criteria

Criterion	What judges look for	Score
Gameplay & Fun Factor	Is it genuinely engaging to play (tension, bluffing, pacing), not just functionally complete?	30 points
Execution Quality	Polish, stability, and responsiveness during the live demo. A smaller, well-executed game beats an ambitious, buggy one.	30 points
Creativity & Vision	Original twists on the Mafia formula, interesting role design, or clever use of the terminal medium.	25 points
Technical Craft	Code structure, handling of hidden per-player state, and quality of the terminal UI/UX under text constraints.	15 points

Demo format: Each team runs a live 3–5 minute play session with real players (teammates or judges), showing at least one full phase cycle (discussion → vote → elimination).